Tomato Temperature Guide

Best temperatures by growth stage, heat and cold stress, and how to protect your plants

Quick answer: Tomatoes are warm-season plants. They grow best at about **21-27 °C by day** and **15-20 °C at night**. Growth slows below about 10-12 °C, and flowers and pollen fail when it is above about 32-35 °C by day or about 24 °C at night. Frost (0 °C) kills the plant.

1. Best temperature by growth stage

Stage	Best range	Notes
Seed germination	21-27 °C (soil)	Possible from about 15 °C but slow. Above 35 °C germination is poor.
Seedlings	Day 18-24 °C, night 15-18 °C	Cold, wet soil causes slow growth and root problems.
Transplanting	Soil at least 15-16 °C	Wait until the cold nights have passed.
Vegetative growth	Day 21-27 °C, night 15-20 °C	Good balance of leaf and root growth.
Flowering and fruit set	Day 21-29 °C, night 13-21 °C	The most sensitive stage. See section 2.
Fruit ripening	18-24 °C	Red color is best in this range.
Harvesting and storage	Pick in cool hours. Store at 10-15 °C	See section 5. Avoid storing below 10 °C.

2. Fruit set: the critical limits

- **Too cold:** night temperatures below about 10-13 °C give poor pollination, small or deformed fruit (catfacing) and few seeds.
- **Too hot:** day temperatures above about 32-35 °C, or night temperatures above about 24 °C, make pollen sterile. Flowers drop and no fruit forms.
- High humidity or very dry air at the same time makes pollination worse.

3. Cold stress

Temperature	What happens
Below 15 °C for long periods	Slow growth, poor nutrient uptake (especially phosphorus), leaves may turn yellow or purple.
Below 10 °C	Chilling injury: growth almost stops, fruit does not ripen, flowers fail.
Around 0 °C or below	Frost. Leaves and stems turn black and die.

What to do: use low tunnels, row covers or a greenhouse; plant later or choose cold-tolerant varieties; use black plastic mulch to warm the soil; water in the morning so the soil keeps its heat; avoid cold, wet soil.

4. Heat stress

Temperature	What happens
Above about 30 °C	Lycopene (red color) forms poorly, so fruit can stay yellow or orange. Plants use more water.
Above about 32-35 °C	Flower drop, no fruit set, leaf curling, sunscald on fruit, cracking.
Heat with uneven watering	More blossom end rot (calcium problems) and yellow leaves.

What to do: use 30-40% shade net in the hottest months; ventilate greenhouses and use fans; mulch and water deeply and regularly; do not remove too many leaves (they shade the fruit); choose heat-tolerant varieties; plant so flowering does not fall in the hottest weeks.

5. Harvesting stage

Fruit usually ripens about 6 to 8 weeks after fruit set. This is faster in warm weather and slower in cool weather, and it depends on the variety.

Ripeness stage	Color	Typical use
Mature green	Fully green, glossy, full size	Can ripen after harvest. Used for long transport.
Breaker	First pink or red on less than 10% of the fruit	Good for shipping to markets.
Turning / Pink	10% to 60% pink or red	Good for nearby markets.
Light red	60% to 90% red	Best for local sale and home use.
Red (ripe)	More than 90% red	Best flavor. Pick and use quickly.

Harvest and storage temperature	What to know
Ripening after harvest: 18-24 °C	Ripens well and develops good color. Above 30 °C color is poor, and below about 12 °C ripening is very slow.
Storage of mature green fruit: 12-15 °C	Slows ripening without damage. Keep humidity high (about 90-95%).
Storage of ripe fruit: 10-13 °C	Good for a few days. Do not keep fruit below 10 °C for long. Chilling injury causes watery, pitted fruit with poor flavor.
Do not refrigerate unripe fruit	Cold stops ripening and damages the fruit. Ripe fruit can go in the fridge for a short time only, but flavor drops.

- **Pick in the morning**, when the fruit is cool. Hot fruit (above about 30 °C) spoils faster.
- In hot weather, pick every 2-3 days and keep the harvested fruit in shade.
- Pick dry fruit. Wet fruit is more likely to rot.
- Cold nights or frost coming: harvest all mature green fruit and ripen it indoors at 18-24 °C.
- Heat can cause sunscald and uneven color. Cool, wet weather can cause cracking and rot, so harvest earlier at the breaker stage.

6. Temperature and diseases or pests

Problem	Weather it likes
Late blight	Cool and wet, 10-24 °C
Septoria leaf spot	Warm and wet, 20-25 °C
Leaf mold	22-25 °C with very high humidity (above 85%)
Early blight	Warm, 24-29 °C, with wet and dry days

Problem	Weather it likes
Bacterial spot	Warm and wet, 24-30 °C
Target spot	Warm and humid, 20-30 °C
Spider mites	Hot and dry, above about 27 °C
Yellow Leaf Curl Virus (whiteflies)	Warm weather, when whitefly numbers grow fast

Knowing the weather helps you act early, for example spraying or opening vents before the disease arrives.

7. Temperature and yellow leaves

- **Cold:** leaves turn yellow or purple-green because the roots cannot take up nutrients well. It improves when it warms up.
- **Heat:** leaves curl, yellow and dry, often with flower drop. Check water and shade.
- Always compare with other causes (water, nutrition, pests, disease) before you decide that temperature is the reason.

8. Simple checklist

- Measure air temperature (day and night) and soil temperature, not only the weather forecast.
- Keep nights between 13 and 21 °C at flowering if you can.
- Protect from frost and cold nights. Provide shade and ventilation in hot weather.
- Water regularly, because heat and uneven watering together cause fruit problems.
- Choose varieties that fit your climate (heat-tolerant or cold-tolerant).
- Harvest in the morning, ripen at 18-24 °C, and store at 10-15 °C (not colder).

Note: The numbers are typical ranges. Exact limits depend on the variety, humidity, light and plant age. Ask a local agronomist for advice that fits your region.